DAVID K. TAN

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CLEARANCE

Active TS Clearance

SKILLS

- Programming: C++, Python, Java, Bash.
- Software Engineering: Object-Oriented Design, Software Architecture, Agile/Scrum.
- DevSecOps: GitLab, CI/CD, Git, Jenkins, Kubernetes, Docker.
- Platforms and Cloud: Linux/Unix, AWS, Red Hat Enterprise Linux (RHEL).
- Quality: Unit/Integration Testing, Test Automation, Static Analysis, Debugging.
- Systems Engineering: Systems Integration and Verification & Validation.

PROFESSIONAL EXPERIENCE

Software Design Engineer III, **Torch Technologies**, Mar 2026 - Current
- Develop C++ and Python software across multi-component defense software applications, taking changes from requirements through testing to confirm the components work together.
- Diagnose and fix complex software problems in Linux using debugging tools, Git, and automated tests, reducing recurring defects by approximately 10%.
- Add automated C++ tests to check required behavior and catch problems earlier when software changes.
- Support CI/CD and software delivery workflows using GitLab, containerized environments, and DevSecOps practices to improve build and test repeatability.
- Improve C++ code and integration workflows, completing approximately 10% more simulation work per unit of time.

Software Engineer II, **BAE Systems** , Sep 2024 - Jan 2026, Utah
- Developed and tested C++ flight simulation software for the LGM-35A Sentinel program, linking every software change to its required flight-performance criteria.
- Investigated unexpected results and corrected C++ flight models to improve accuracy and reduce later verification work.
- Implemented regression tests and reviewed code against requirements to keep updated models reliable.

Software Engineer I, **Space Dynamics Laboratory**, Jan 2023 - Sep 2024, Utah
- Fixed defects and improved C++ and Python software in FORGE MDPAF to keep production data processing reliable.
- Contributed software changes across shared modules and interfaces using version control, debugging, testing, and collaborative development practices.

Software Engineer Intern, **Bowie State University**, Jun 2020 - Aug 2022
- Designed and simulated network attacks, improving threat-detection effectiveness by 28%

PROJECTS

- Mission Planning Software: Designed and built a full-stack mission-planning application with role-based workflows, REST APIs, and audit logging, combining software engineering with operational mission needs and reducing planning data-entry errors by 30%.
- Missile Flight Dynamics Simulation: Developed a standalone C++ 6-DOF missile simulation with guidance, autopilot control, and flight-dynamics modeling; evaluated system behavior and performance through Monte Carlo analysis.

EDUCATION

Master of Science, Computer Science — May 2027
Georgia Institute of Technology

Bachelor of Science, Computer Science — Dec 2022
Valdosta State University

CERTIFICATIONS

CISSP | RHCSA | CKA | CKS | INCOSE CSEP | Security+

PUBLICATION

- Co-authored peer-reviewed Springer publication on distributed fault-containment simulation using randomized scheduling, published in *Advanced Communication and Intelligent Systems.*